

QUESTION 1 (15 marks) Start on a new page

Marks

(a) Consider the complex numbers $z_1 = \sqrt{2}(1 + i\sqrt{3})$ and $z_2 = 2\sqrt{6}(1 + i)$

i) Express $z = \frac{z_1}{z_2}$ exactly in the form $x + iy$ where x and y are real numbers. 2

ii) Write z_1, z_2 and z in modulus/argument form. 2

iii) Hence find the exact value of $\cos \frac{\pi}{12}$. 1

iv) On an Argand diagram draw the vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ to represent z_1, z_2 and $z_1 - z_2$ respectively 2

b) If $z = x + iy$ show that 2

$$z + \frac{|z|^2}{z} = 2\operatorname{Re}(z)$$

c) By applying De Moivre's theorem and also by expanding 4

$(\cos \theta + i \sin \theta)^4$, express $\cos 4\theta$ in terms of $\cos \theta$

(d) If z is a complex number such that 2

$z = k(\cos \theta + i \sin \theta)$, where k is real, show that

$$\arg(z + k) = \frac{\theta}{2}$$

QUESTION 2 (15 marks) Start on a new page

Marks

- (a) i) By solving the equation $z^3 = 1$ find the three cube roots of one. 2
- ii) Let ω be a cube root of one where ω is not real. Show that $1 + \omega + \omega^2 = 0$ 1
- iii) Find the quadratic equation with integer coefficients that has roots $(4 + \omega)$ and $(4 + \omega^2)$ 3
- iv) Given $x = a + b$ 2
- $$y = a\omega + b\omega^2$$
- and $z = a\omega^2 + b\omega$ prove that
- $$x^2 + y^2 + z^2 = 6ab$$
- (b) i) Show that $\frac{(1 + i\sqrt{3})^6}{(\sqrt{3} - i)^k} = 2^{6-k} \left(\cos \frac{k\pi}{6} + i \sin \frac{k\pi}{6} \right)$ 2
- ii) For what value of k is $\frac{(1 + i\sqrt{3})^6}{(\sqrt{3} - i)^k}$ purely imaginary. 1
- c) Let OABC be a square in the Argand diagram where O is the origin. The points A and C represent the complex numbers z and iz respectively.
- i) Find the complex number represented by B. 1
- ii) The square OABC is now rotated through 45° in an anticlockwise direction about O to $OA'B'C'$. Find the complex numbers A', B' and C' . 3

QUESTION 3 (15 marks) Start on a new page

Marks

- (a) Find the square root of $5-12i$ and hence solve the equation

4

$$z^2 + 4z - 1 + 12i = 0$$

- (b) Find the locus of z if

3

$$\frac{3z+i}{z-2} \text{ is purely imaginary.}$$

- (c) Indicate on an Argand diagram the regions representing.

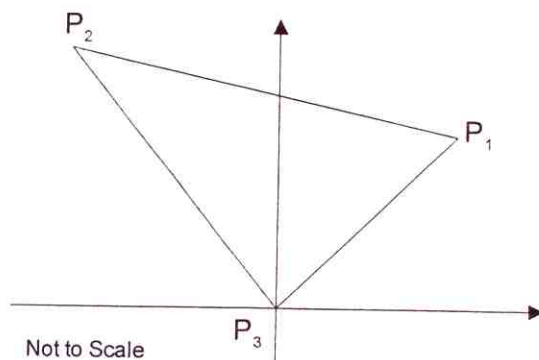
i) $\operatorname{Re}(z + iz) \geq 2$

2

ii) $1 \leq |z - 1 - i| \leq 3$ where $z = x + iy$

2

- (d)



The points P_1, P_2 and P_3 represent the complex numbers z_1, z_2 and z_3 respectively. (NOTE: $z_3 = 0$)

- i) If P_1, P_2 and P_3 are the vertices of an equilateral triangle, show that

2

$$\frac{z_2}{z_1} = \frac{1 + i\sqrt{3}}{2} \text{ and deduce that } z_1^2 + z_2^2 = z_1 z_2$$

- ii) Deduce that if z_1, z_2 and z_3 are ANY three complex numbers at the vertices of an equilateral triangle then

2

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_1 z_3$$

END OF EXAMINATION

$$a) i) z = \frac{\sqrt{2}(1+i\sqrt{3})}{2\sqrt{6}(1+i)} \times \frac{(1-i)}{(1-i)}$$

$$= \frac{1}{2\sqrt{3}} \frac{(1+\sqrt{3}) + (\sqrt{3}-1)i}{2}$$

$$= \frac{1+\sqrt{3}}{4\sqrt{3}} + \frac{(\sqrt{3}-1)i}{4\sqrt{3}}$$

$$ii) z_1 = 2\sqrt{2} \left(\frac{1}{2} + i\frac{\sqrt{3}}{2} \right)$$

$$= 2\sqrt{2} (\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$z_2 = 2\sqrt{6} \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$$

$$= 4\sqrt{3} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$$

$$z = \frac{2\sqrt{2}}{4\sqrt{3}} \cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$$

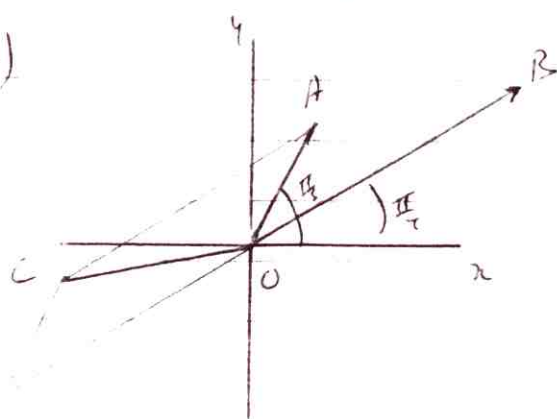
$$= \frac{\sqrt{2}}{2\sqrt{3}} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

$$iii) \frac{\sqrt{2}}{2\sqrt{3}} \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{4\sqrt{3}}$$

$$\cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{4\sqrt{3}} \times \frac{2\sqrt{3}}{\sqrt{2}}$$

$$= \frac{1+\sqrt{3}}{2\sqrt{2}}$$

b) iv)



$$b) z = x + iy$$

$$\therefore z + \frac{|z|^2}{3} = x + iy + \frac{x^2+y^2}{3} \times \frac{x-iy}{x-iy}$$

$$= x + iy + \frac{(x^2+y^2)}{3} \cdot \frac{x-iy}{x^2+y^2}$$

$$= x + iy + x - iy$$

$$= 2x$$

$$= 2 \operatorname{Re}(z)$$

$$c) \cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$$

$$\cos 4\theta = \operatorname{Re}(\cos \theta + i \sin \theta)^4$$

$$= \cos^4 \theta + 6i^2 \cos^2 \theta \sin^2 \theta + i^4 \sin^4 \theta$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

$$d) z = k(\cos \theta + i \sin \theta)$$

$$z+k = k(\cos \theta + i \sin \theta) + k$$

$$= k(1 + \cos \theta + i \sin \theta)$$

$$= k \left(1 + 2 \cos^2 \frac{\theta}{2} - 1 + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$$

$$= 2k \left(\cos^2 \frac{\theta}{2} + i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$$

$$z+k = 2k \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$$

$$\arg(z+k) = \frac{\theta}{2}$$

2) a) i) $z^3 = 1$

let $z = r \cos \theta$

$z^3 = r^3 \cos 3\theta = 1$

$|z^3| = 1 \therefore r = 1$

$\arg z^3 = 0$

$\therefore 3\theta = 0, 2\pi, 4\pi$

$\theta = 0, 2\pi/3, -2\pi/3$

$z = 1, \cos 2\pi/3, \cos -2\pi/3$

$= 1, -\frac{1+i\sqrt{3}}{2}, -\frac{1-i\sqrt{3}}{2}$

ii) $z^3 = 1$

$z^3 - 1 = 0$

$(z-1)(z^2+z+1) = 0$

if ω is not real $\omega \neq 1$ a/c - 1)

$\therefore \omega^2 + \omega + 1 = 0$

OR $z^3 - 1 = 0$

$\Sigma d = -\frac{b}{a} = 0 \therefore 1\omega + \omega^2 = 0$

after showing ω^3 is root.

iii) $\alpha + \beta = 4 + \omega + 4 + \omega^2$

$= 8 + \omega + \omega^2$

$= 7 + (1 + \omega + \omega^2)$

$= 7$

$\alpha\beta = (4 + \omega)(4 + \omega^2)$

$= 16 + 4\omega^2 + 4\omega + \omega^3$

$= 13 + (4 + 4\omega + 4\omega^2)$

$= 13$

Quadratic Eqn is $x^2 - 7x + 13 = 0$.

iv) $x^2 + y^2 + z^2 = (a+b)^2 + (a\omega + b\omega^2)^2 + (a\omega^2 + b\omega)^2$

$= a^2 + 2ab + b^2 + a^2\omega^2 + 2ab\omega^3 + b^2\omega^4 + a^2\omega^4 + ab\omega^3 + b^2\omega^2$

$= a^2 + 2ab + b^2 + a^2\omega^2 + 2ab + b^2\omega + a^2\omega + 2ab + b^2\omega^2$

$= a^2 + b^2 + 6ab + \omega(a^2 + b^2) + \omega^2(a^2 + b^2)$

$= a^2 + b^2 + 6ab + (a^2 + b^2)(\omega + \omega^2)$ but $\omega + \omega^2 = -1$

$= 6ab$

b) i) $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^6} = \frac{2^6 \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^6}{2^6 \left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)^6}$

$= \frac{2^{6-k} \cos 6 \times \frac{\pi}{6}}{\cos k \times -\frac{\pi}{6}}$

$= 2^{6-k} \cos(2\pi + \frac{k\pi}{6})$

$= 2^{6-k} \left(\cos(2\pi + \frac{k\pi}{6}) + i \sin(2\pi + \frac{k\pi}{6}) \right)$

$= 2^{6-k} \left(\cos \frac{k\pi}{6} + i \sin \frac{k\pi}{6} \right)$

ii) purely imaginary w

$\frac{k\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}$

$k = \pm 3\pi \pm 15\pi$

$= \pm 3(4n+1) \quad n=0,1,2$

OR SIMILAR.

c) i) $z_2 = z + iz = (1+i)z$

ii) $A^1 \Rightarrow z_1 = z \cos \frac{\pi}{6}$

$= \frac{z}{\sqrt{2}} (1+i)$

$A^2 \Rightarrow z_2 = (1+i)z_1 \cos \frac{\pi}{6}$

$= z (1+i) \left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} \right)$

$= \frac{z}{\sqrt{2}} \cdot 2i = \frac{2iz}{\sqrt{2}} = \sqrt{2}iz$

$A^3 \Rightarrow z_3 = i z_2$

$= z i (1+i)$

$= \frac{\sqrt{2}}{2} z (-1+i)$

3) a) $z^2 = 5 - 12i$ let $z = x + iy$

$$x^2 - y^2 + 2ixy = 5 - 12i$$

$$x^2 - y^2 = 5$$

$$xy = -6$$

$$\text{let } y = \frac{-6}{x}$$

$$x^2 - \frac{36}{x^2} = 5$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2 - 9)(x^2 + 4) = 0$$

$$x = \pm 3 \quad y = \mp 2$$

$$\sqrt{5-12i} = \pm(3-2i) \quad *$$

$$z^2 + 4z - 1 + 12i = 0$$

$$z = \frac{-4 \pm \sqrt{16 - 4(12i-1)}}{2}$$

$$= \frac{-2 \pm \sqrt{20 - 48i}}{2}$$

$$= -2 \pm \sqrt{5-12i}$$

$$= -2 \pm 2(3-2i)$$

$$= -8+4i \text{ or } 4-4i$$

b) let $z = x + iy$

$$\frac{3z+i}{z-2} = \frac{3(x+iy)+i}{x+iy-2}$$

$$= \frac{3x + (3y+1)i}{(x-2) + iy} \cdot \frac{(x-2) - iy}{(x-2) - iy}$$

$$R(z) = \frac{3x(x-2) + y(3y+1)}{(x-2)^2 + y^2}$$

$$= 0$$

$$3x^2 - 6x + 3y^2 + y = 0$$

$$x^2 - 2x + 1 + y^2 + \frac{y}{3} + \frac{1}{36} = \frac{37}{36}$$

$$(x-1)^2 + \left(y + \frac{1}{6}\right)^2 = \left(\sqrt{\frac{37}{36}}\right)^2$$

$$\therefore \text{circle centre } \left(1, -\frac{1}{6}\right) \quad r = \frac{\sqrt{37}}{6}$$

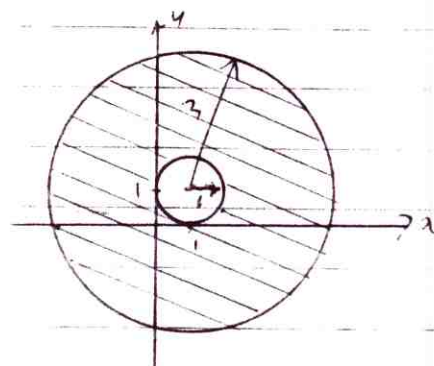
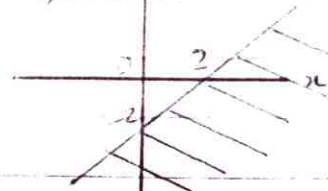
$$\text{including } x=2 \quad y=0$$

c) let $z = x + iy \quad \therefore z + iz$

$$z + iz = (x+iy) + xi - y$$

$$= (x-y) + (x+iy)i$$

$$x-y \geq 2$$



d) 1) $z_2 = z_1 \cos \frac{\pi}{3}$

$$\frac{z_2}{z_1} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$= \frac{1 + i\sqrt{3}}{2} \quad *$$

2) $z_2 = z_1 \cos \frac{\pi}{3}$

$$z_1^2 + z_2^2 = z_1^2 + z_1^2 \cos 2\frac{\pi}{3}$$

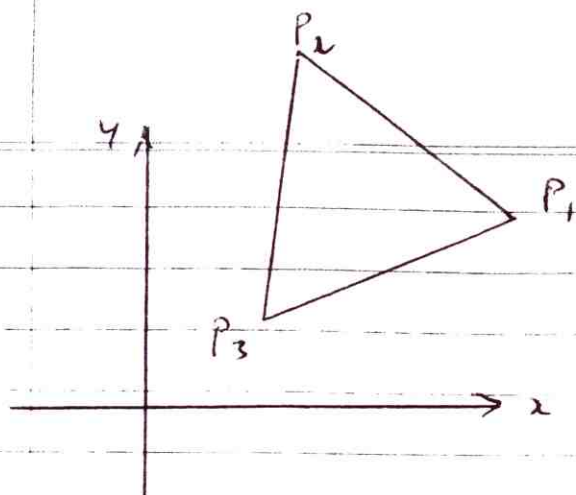
$$= z_1^2 + z_1^2 \left(-\frac{1}{2} + i\sqrt{\frac{3}{2}}\right)$$

$$= z_1 \left(z_1 + z_1 \left(-\frac{1}{2} + i\sqrt{\frac{3}{2}}\right)\right)$$

$$= z_1 \left(1 - \frac{1}{2} + i\sqrt{\frac{3}{2}}\right) z_1$$

$$= z_1 \left(\frac{1}{2} + i\sqrt{\frac{3}{2}}\right) z_1$$

$$= z_1 z_2$$



ii) from $z_1^2 + z_2^2 = z_1 z_2$
 when $\triangle$ has P_3 at 0

$$z_1 \rightarrow z_1 - z_3$$

$$z_2 \rightarrow z_2 - z_3$$

when P_3 not at 0

$$(z_1 - z_3)^2 + (z_2 - z_3)^2 = (z_1 - z_3)(z_2 - z_3)$$

$$z_1^2 - 2z_1 z_3 + z_3^2 + z_2^2 - 2z_2 z_3 + z_3^2 = z_1 z_2 - z_1 z_3 - z_2 z_3 + z_3^2$$

$$\therefore z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_1 z_3$$